

The Equilibrium Effects of Vacancy Referrals

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Abstract

Job vacancy referrals (VRs) are used by public employment services to connect unemployed workers and firms that otherwise would not have matched. Existing quasi-experimental evidence shows that VRs increase the job-finding probabilities of VR recipients. Their impact on economywide employment, however, depends on the magnitude of equilibrium effects. We develop and estimate a search and matching model that incorporates several sources of equilibrium effects: congestion in the hiring process, endogenous worker search effort, and firms' vacancy posting responses. We estimate the model using German data on unemployed workers' job search behavior and firms' hiring decisions. We find that equilibrium effects substantially attenuate the impact of VRs: assigning the status-quo VR policy to an individual worker increases their matching probability by 6.9 percentage points, while implementing the same policy economywide increases the aggregate worker matching rate by only 2.3 percentage points. Targeting can substantially increase the effectiveness of VRs. Reallocating VRs from workers with low application costs toward those with high application costs, while keeping the overall number of referrals fixed, increases the worker matching rate by 4 percentage points.

Keywords: unemployment, vacancy referrals, wage, unemployment insurance, monitoring, moral hazard, structural estimation, counterfactual policy evaluation

JEL classification: J64, J65, C51, C54

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1 Introduction

In most OECD countries, public employment services (PES) provide job search assistance to unemployment benefit recipients. A common component of job search assistance is the use of vacancy referrals (VRs), which are directives to apply for specific job openings. PES use VRs both to support benefit recipients' own job search efforts and to encourage individuals whose search effort is deemed insufficient to apply for additional jobs (see, e.g., Grubb, 2001; Immervoll and Knotz, 2018). The overarching motivation for using VRs as part of active labor market policy (ALMP) is to bring together unemployed workers and firms that otherwise would not have matched.

An open question in evaluating VRs is the extent to which their impact is attenuated by equilibrium effects.¹ In this paper, we study three channels through which equilibrium effects may arise. First, we examine whether assigning VRs to a group of unemployed workers reduces the job-finding prospects of other unemployed workers. Second, we analyze how VRs affect unemployed workers' own job search effort. Third, we study how firms' vacancy posting behavior responds to changes in the use of VRs.

Existing empirical evidence suggests that VRs have a positive impact on the job-finding probabilities of VR recipients (see, e.g., Bollens and Cockx, 2017; van den Berg, Hofmann, and Uhlendorff, 2014).² Further evidence is provided by Fougère, Pradel, and Roger (2009), who estimate a partial equilibrium job search model to study the extent to which VRs disincentivize job search effort. They find moderate disincentive effects, but show that the net employment effect of VRs remains positive and large in magnitude.

Our main goal in this paper is to investigate the extent to which the positive effects of VRs on individual job finding probabilities translate into an increase in the economywide employment rate, and to quantify and characterize the forces that may mitigate the translation. A second goal is to understand how policies should be designed to avoid exerting offsetting equilibrium effects. To this end, we develop and estimate a dynamic model that accounts for several channels through which equilibrium effects may operate.

Our model reflects the fact that unemployed workers submit multiple job applications, while typically searching for only one job at a time. Similarly, firms face a queue of multiple applica-

¹See Abbring and Heckman (2007) for a survey of the literature on general equilibrium evaluation of ALMPs.

²Using data from Flanders, Bollens and Cockx (2017) find that VRs substantially increase job-finding probabilities in both the short and the long run. Using German data, van den Berg, Hofmann, and Uhlendorff (2014) show that VRs are associated with higher job-finding probabilities. Using data from Sweden, Engström, Hesselius, and Holmlund (2012) find that only about one third of VRs result in job applications, but that intensified monitoring increases the share of VRs that lead to applications.

tions for a given vacancy. When a worker receives a VR and applies, an additional application is added to the queue for that vacancy, which typically reduces the hiring probability of all other applicants. We refer to this mechanism as congestion in the hiring process. Such congestion may result in crowding out, whereby a referred applicant displaces another applicant who would otherwise have been hired. As a result, the net effect of VRs on the economywide employment rate may be zero if a VR simply crowds out another applicant who, as a consequence, remains unmatched. At the other extreme, VRs may lead to the filling of vacancies that would otherwise have remained unfilled, either because no other workers applied or because displaced applicants find employment elsewhere. Our structural model formalizes this reasoning and allows us to quantify the extent of congestion and the resulting crowding-out effects. In addition, the model accounts for endogenous changes in firms' vacancy posting behavior in response to VRs, as well as for potential disincentive effects of VRs on workers' own job search effort, as in Fougère, Pradel, and Roger (2009).

More specifically, we develop a dynamic search and matching model in which unemployed workers and firms match at random according to an urn-ball matching technology (see, e.g., Rogerson, Shimer, and Wright (2005) and Petrongolo and Pissarides (2001)). Workers send out multiple applications to a (random) set of firms, and each firm hires at most one worker from its applicant pool.³ Firms optimally choose whether to post a vacancy and which wage to post. They take the resulting applicant pool as given and select one worker from it when filling the vacancy.

In each period, unemployed workers choose how many applications to send to firms and may additionally receive VRs from the PES. Receiving VRs increases the job-finding probability of the worker who receives it, but it also increases congestion in the applicant pool of the targeted vacancy, thereby reducing the hiring probabilities of other applicants. The resulting degree of crowding out depends on labor market tightness and on the number of applications sent by the VR recipient and by other workers applying to the same vacancy. In addition to congestion effects, changes in the number of VRs in the economy affect equilibrium outcomes by influencing firms' vacancy and wage posting decisions as well as workers' incentives to submit regular applications.

We estimate the model using the generalized method of moments (GMM), targeting moments from German survey microdata on unemployed workers' search behavior and post-unemployment outcomes, as well as from data on firms' hiring decisions. The worker-side data are informative,

³Our model builds on the search and matching framework with multiple applications developed in Albrecht et al. (2004), Albrecht, Gautier, and Vroman (2006), and Gautier, Moraga-González, and Wolthoff (2016).

among other things, about the number of applications sent and VRs received for an inflow sample into unemployment in 2008. The firm-side data contain information on the number of hires and unsuccessful searches in the previous year, as well as detailed information on the applicant pool for the firm’s most recent job opening, including the number of candidates referred through VRs and whether the hired candidate was referred through a VR. We allow for worker and firm heterogeneity by estimating the model separately for four labor market submarkets, defined by region (East versus West Germany) and sector (manufacturing versus services).

We use the estimated model to conduct a series of counterfactual policy experiments. We first scale the use of VRs up and down relative to the status-quo policy and study their effects on worker search behavior, congestion, vacancy posting, wages, and worker matching rates. We then compare the effects of VRs with those of alternative labor market policies and study the gains from targeting VRs toward workers with high search costs. Finally, we quantify the equilibrium effects of VRs by comparing the individual-level effect of a policy that leaves equilibrium conditions unchanged with its effect when the same policy is implemented at scale.

We find that expanding VRs increases the total volume of regular applications and VRs in the labor market and raises worker matching rates. Workers who receive more VRs also submit more regular applications, reflecting complementarities between VRs and workers’ own search activity. The resulting increase in contacts between workers and vacancies raises congestion at vacancies. Despite this increase in congestion, worker matching rates increase, unemployment falls, and vacancy posting rises, while wages increase modestly. The effects of VRs are qualitatively similar to those of policies that reduce application costs, such as job search assistance.

Importantly, our results also show that targeting can substantially increase the effectiveness of VRs. We find that a policy that reallocates VRs from unemployed workers with low search costs toward those with higher search costs, while holding the overall volume of VRs fixed, increases the worker matching rate by 4 percentage points (15%). Thus, targeting VRs toward workers who would otherwise search relatively little can substantially improve the effectiveness of the policy without requiring an increase in the overall number of referrals.

While these results show that VRs do increase worker matching rates, we find that much of the potential effect remains unrealized due to equilibrium effects that substantially attenuate the effects of VRs. To quantify their magnitude, we consider the status-quo VR policy and compare assigning this policy to a single unemployed worker with rolling out the same policy to the entire unemployed population. Assigning the policy to an individual worker increases the matching rate of this worker by 6.9 percentage points, whereas the economywide roll-out raises the aggregate worker matching rate by only 2.3 percentage points. Thus, the effect in

the absence of equilibrium adjustments is three times as large as the effect when the policy is implemented at scale. These results suggest that estimates from small-scale quasi-experimental evaluations that neglect equilibrium effects may substantially overstate the effects of VRs when implemented at scale.

The contribution of this paper is twofold. First, we develop a structural model of the labor market that incorporates VRs and accounts for three main channels through which equilibrium effects may operate: congestion effects in the hiring process, changes in workers' own search effort, and endogenous changes in firms' vacancy posting behavior. Second, we estimate the model using worker- and firm-level data and use the estimated model to quantify the discrepancy between the individual-level effects of VRs documented in previous empirical studies and the effects of VR policies when implemented at scale.

We thereby contribute to a small literature that studies the equilibrium effects of ALMPs. In particular, Crépon et al. (2013) and Gautier et al. (2018) provide empirical evidence on the equilibrium effects of ALMPs.⁴ Crépon et al. (2013) interpret their empirical findings through the lens of a stylized search and matching model in which equilibrium effects arise exclusively from firms' vacancy posting responses to changes in ALMPs. Gautier et al. (2018) estimate a structural model to analyze counterfactual policy scenarios in which search costs are reduced. Relative to these papers, we provide the first study of the equilibrium effects of VRs and the channels through which these equilibrium effects operate.

The congestion effects we study are related to congestion on online dating platforms, where users send many essentially costless signals (Lehmann, Terrier, and Lalive, 2023). A key difference in labor markets is that only workers send applications, whereas firms post vacancies and select applicants.

The remainder of the paper is organized as follows. Section 2 describes the institutional background. Section 3 develops the model, Section 4 describes the data, and Section 5 outlines the model estimation. Section 6 presents the results from the policy simulations. Section 7 concludes.

2 Institutional Background

In most OECD countries recipients of unemployment insurance benefits are subject to a range of job search monitoring and job search assistance practices (see Grubb, 2001). VRs are one of the most important of these practices (see Immervoll and Knotz, 2018). By making use of VRs

⁴In these papers, the ALMPs considered include job search assistance, training programs, as well as VRs.

unemployment insurance agencies seek to support individual benefit recipients' own job search effort, but also to activate benefit recipients whose own search effort is deemed insufficient.

In Germany UI benefit recipients may receive VRs starting immediately after they start claiming benefits. Caseworkers at the public employment services (*Bundesagentur für Arbeit*) select job vacancies from the agency's database and matches them to UI benefit recipients.⁵ The selected benefit recipients are then notified by regular mail about the vacancy, are typically given a brief description about the job opening and are asked to apply for the referred position(s) (this description typically does not include the wage).⁶ Not applying for a referred vacancy or rejecting a resulting job offer may lead to sanction (a temporary benefit reduction), although enforcement of such sanctions is infrequent (see van den Berg, Foerster, and Uhlendorff, 2019).

There are two standard procedures by which job openings may enter as vacancies into the database of the PES. Firms may directly contact the PES and request that a specific vacancy is used for making VRs. Moreover, to find suitable vacancies caseworkers browse public job advertisements such as newspaper- and online-ads, i.e., in principal most publicly advertised job openings may end up being used by the PES for making VRs. There are two restrictions that caseworkers need to abide when making a VR. The job has to be within 2.5 hours commuting distance and has to pay at least the same amount that the benefit recipient receives in UI benefits. There are no other restrictions on the type of jobs that a UI benefit recipient may receive as VRs. In particular job openings referred as VRs do not need to match the pre-unemployment occupation of the UI benefit recipient who is assigned the vacancy.

3 Model

Our model is a dynamic search and matching model in which unemployed workers submit multiple applications in each period and firms post vacancies and wages. Applications are allocated randomly across vacancies. Each firm with at least one applicant selects one applicant at random and makes a job offer at its posted wage. Workers who receive multiple offers choose the most attractive one or may reject all offers in favor of remaining unemployed. VRs are additional applications that unemployed workers do not choose but receive exogenously. Like regular applications, VRs are allocated randomly across vacancies. A VR policy is characterized by the rates at which workers receive VRs.

The model captures several channels through which VRs can generate equilibrium effects.

⁵The same vacancy may be referred to several UI benefit recipients and benefit recipients sometimes receive several VRs at the same time.

⁶A VR typically contains information about the occupation, the working hours and the date of potential job start (see van den Berg, Hofmann, and Uhlendorff, 2014).

First, a VR increases the matching probability of the worker who receives it, but also increases congestion at the targeted vacancy and thereby reduces the matching probabilities of other applicants. Second, VRs may affect workers' own search effort by changing their incentives to submit regular applications. Third, firms respond to changes in the number of applications and VRs through their vacancy posting and wage posting decisions.

In the empirical implementation, we allow model parameters to differ across four submarkets defined by region and sector, as described in Section 5. Here we present the model for a single labor market, suppressing submarket subscripts throughout this section.

Setup Our model is set in discrete time and the time horizon is infinite. There is a unit mass of workers in the economy and a measure v of vacancies, which is endogenously determined. All agents have linear preferences and discount the future at rate β . Firms are homogeneous and workers are heterogeneous in their costs of submitting applications, but are homogeneous in their productivity, i.e., whenever a firm and a worker are matched they produce a constant per period output y .

Hiring process In each time period firms receive multiple applications from different workers. We assume that applications are undirected, i.e., each firm is equally likely to receive a given application. Denote by $p_{\alpha,\iota}$ the population share of workers submitting $\alpha \in \{0, 1, 2, \dots, A\}$ applications and receiving $\iota \in \{0, 1, 2, \dots, J\}$ VRs. The queue length, i.e., the expected total number of regular applications and VRs for a given firm is

$$\begin{aligned}\lambda &= \frac{u}{v} \sum_{\iota=0}^J \sum_{\alpha=0}^A (\alpha + \iota) p_{\alpha,\iota} \\ &= \frac{u}{v} \sum_{\iota=0}^J p_{\iota} \sum_{\alpha=0}^A (\alpha + \iota) p_{\alpha|\iota}.\end{aligned}\tag{1}$$

where p_{ι} denotes the share of unemployed workers receiving ι VRs and $p_{\alpha|\iota}$ denotes the share submitting α regular applications conditional on receiving ι VRs.

As workers are homogeneous in their productivity, firms select an applicant at random. Albrecht et al. (2004) show that in this case the probability that a given application is successful is⁷

$$\psi = \frac{1}{\lambda} (1 - e^{-\lambda}).$$

⁷For a detailed exposition of urn-ball matching with multiple applications see Albrecht, Gautier, and Vroman (2006).

For a worker who submits α applications and receives ι VRs the the probability to receive $k \in \{0, \dots, \alpha + \iota\}$ job offers then is given by

$$\mu(k|\alpha + \iota) = \binom{\alpha + \iota}{k} \psi^k (1 - \psi)^{\alpha + \iota - k}.$$

Workers When unemployed, workers receive unemployment benefits b per period. In each period, they choose how many applications to submit and, upon receiving job offers, whether to accept or reject them. The probability of receiving k job offers, $\mu(k|\alpha + \iota)$, depends on the sum of the number of regular applications submitted, α , and VRs received, ι .

Upon submitting α regular applications and receiving ι VRs the unemployed worker incurs application costs $C(\alpha, \iota) = c \cdot (\max\{\alpha + \phi\iota, 0\})^2$. The parameter c , which scales application costs, is heterogeneous across workers and is redrawn independently in each period (i.e. for a given worker c is iid across time). The parameter ϕ captures how receiving a VR affects application costs. In particular, $\phi > 0$ implies that responding to VRs increases the marginal cost of submitting regular applications, whereas $\phi < 0$ implies complementarities between VRs and regular applications. Such complementarities may arise, for example, because preparing an application or preparing for an interview for a referred vacancy reduces the cost of performing similar steps for other job applications.

The value of unemployment conditional on the realizations of c and ι , and on submitting α regular applications, is

$$V_{U|c,\iota}(\alpha|c, \iota) = b - C(\alpha, \iota) + \beta \left(\sum_{k=1}^{\alpha+\iota} \mu(k|\alpha + \iota) \int_0^\infty \max\{V_U, V_E(w)\} dF^k(w) + \mu(0|\alpha + \iota) V_U \right) \quad (2)$$

where $F(w)$ is the wage offer distribution that workers take as given, so that $F^k(w)$ is the distribution of the highest wage among k independent offers. For a given worker the optimal number of applications is therefore

$$\alpha(\iota, c) = \operatorname{argmax}_{\alpha} V_{U|c,\iota}(\alpha|c, \iota). \quad (3)$$

The unconditional (ex ante) value of unemployment takes into account workers' optimal application choices and is computed prior to the realizations of the application cost parameter and

the number of VRs, i.e., by taking expectations over c and ι .

$$V_U = \int_0^\infty \sum_{\iota=0}^J p_\iota \max_\alpha V_{U|c,\iota}(\alpha|c, \iota) dS(c) \quad (4)$$

In terms of timing, workers observe both the number of VRs they receive, ι , and their application cost parameter, c , before choosing the number of regular applications to submit.

Employed workers receive a constant wage w each time period for the duration of their employment spell and are laid off at rate δ . The value of employment is therefore

$$V_E(w) = w + \beta [(1 - \delta)V_E(w) + \delta V_U] \quad (5)$$

It is straightforward to show that V_E is strictly increasing in w . A worker's decision rule in accepting and rejecting job offers can therefore be summarized by a reservation wage $\bar{w}$.

Using equations (5) and (4) and rearranging yields the reservation wage equation

$$\bar{w} = b + \int_0^\infty \sum_{\iota=0}^J p_\iota \max_\alpha \left(\beta \sum_{k=1}^{\alpha+\iota} \mu(k|\alpha + \iota) \int_{\bar{w}}^\infty \frac{w - \bar{w}}{1 - \beta(1 - \delta)} dF^k(w) - C(\alpha, \iota) \right) dS(c). \quad (6)$$

Firms Firms in each time period decide whether to post a vacancy and, if they decide to post a vacancy, which wage to post. The value a firm attributes to being matched with a worker and paying wage w is

$$\begin{aligned} V_M(w) &= y - w + \beta(1 - \delta)V_M(w) \\ &= \frac{y - w}{1 - \beta(1 - \delta)}, \end{aligned} \quad (7)$$

where y is the productive value of a match. The value of posting a vacancy is given by

$$V_F(w) = -k + \beta m_F(w) V_M(w),$$

where k is the cost of posting a vacancy and $m_F(w)$ is the endogenous probability for a given firm to be matched with a worker. Gautier, Moraga-González, and Wolthoff (2016) show that the matching probability for a firm is given by

$$m_F(w) = \frac{u}{v} \sum_{k=1}^{A+J} k q_k F^{k-1}(w),$$

where q_k is the population share of unemployed workers receiving k job offers,

$$\begin{aligned} q_k &= \sum_{\iota=0}^J \sum_{\alpha=0}^A \mu(k|\alpha, \iota) p_{\alpha, \iota} \\ &= \sum_{\iota=0}^J p_{\iota} \sum_{\alpha=0}^A \mu(k|\alpha, \iota) p_{\alpha|\iota}. \end{aligned} \quad (8)$$

We assume free entry, i.e. it is optimal for new firms to enter the market and post vacancies until the value of posting a vacancy is driven down to zero, i.e., until $V_F(w) = 0$ for all w or equivalently

$$\begin{aligned} 0 &= -k + \beta m_F(w) V_M(w) \\ &= -k + \beta m_F(w) \frac{y - w}{1 - \beta(1 - \delta)}, \end{aligned} \quad (9)$$

where the second equality is obtained by inserting (7). From a firm's perspective it is never optimal to post wages below the workers reservation wage $\bar{w}$. Denote the smallest wage posted with positive probability by $w_{\min}$. This wage equals the reservation wage, i.e., $w_{\min} = \bar{w}$. The firm posting the smallest wage will only attract unemployed workers who receive no other job offer, i.e., $m_F(\bar{w}) = q_1$. Using $V_F(w) = V_F(\bar{w}) = 0$ can be used together with $m_F(\bar{w}) = q_1$ to characterize the wage offer distribution by

$$\sum_{k=1}^{A+J} k q_k F^{k-1}(w)(y - w) = q_1(y - \bar{w}). \quad (10)$$

For given y , q_k and $\bar{w}$, (10) can be used to solve for the wage offer distribution F pointwise at any w .

Steady State For the economy to be in steady state, flows into unemployment need to equal flows out of unemployment, i.e.,

$$0 = \delta e - m_W u, \quad (11)$$

so that the steady state unemployment is given by $u = \frac{\delta}{\delta + m_W}$.

Equilibrium A steady state equilibrium of the economy is given by a queue length λ , a policy function, $\alpha(\iota, c)$, a reservation wage $\bar{w}$, a measure of vacancies v and a wage offer distribution F , such that (1), (3), (6), (9), (10) and (11) are satisfied.

Distribution of Accepted Wages In the described setup a closed form relationship between the equilibrium wage offer distribution F and the distribution of accepted wages can be derived. Making use of this relationship will allow us to compute moments from the accepted wage distribution analytically. This will prove useful for the structural estimation of the model as it allows us to target moments of the distribution of accepted wages rather than moments of the (unobserved) wage offer distribution. Denote the distribution of accepted wages by $G(w)$. Then in equilibrium it holds that

$$G(w) = \frac{\sum_{k=1}^{A+J} q_k F^k(w)}{1 - q_0}$$

(see Gautier, Moraga-González, and Wolthoff, 2016). We use this relationship in computing moments of the distribution of accepted wages implied by our model by using $F(w)$ to compute $G(w)$ at a set of approximation points and applying numerical integration techniques to obtain the desired moments.

4 Data

We estimate our model using a combination of worker and firm data. For worker-side information we use the IZA Evaluation Dataset, a survey covering a large sample of workers who became unemployed between June 2007 and May 2008 in Germany.⁸ The survey covers a random sample of 17,396 newly unemployed workers aged 16 to 54, who are surveyed up to three times: once as soon as possible after inflow into unemployment, again roughly one year later, and in a final wave approximately three years after the first interview.

The data contain detailed information on individuals' labor market histories and job search behavior, including the number of applications submitted, vacancy referrals received, and monthly unemployment benefits during the last unemployment spell before the survey interview, as well as the start and end dates of all unemployment and employment spells since inflow. In addition, the data include basic sociodemographic characteristics such as age, gender, region of residence, and highest educational attainment.

For information on firm hiring we use the IAB Job Vacancy Survey, a quarterly representative survey among German establishments that collects detailed information on recruiting processes.⁹ The survey is conducted as a repeated cross-section and covers approximately 10,000 to 16,000

⁸See Caliendo et al. (2011) and Arni et al. (2014) for a detailed description of the IZA Evaluation Dataset.

⁹See Kubis (2018) for a detailed description of the IAB Job Vacancy Survey. The unit of observation in the IAB Job Vacancy Survey is the establishment. For simplicity, we refer to establishments as firms throughout the paper.

firms per year starting in 2000.

The data contain detailed information on firms' hiring and recruitment activities, including the number of applications received for the most recently filled vacancy, the number of hires over the past year, as well as the firm's industry sector and region of location.

Sample In selecting our sample we make use of a time window in which our two data sources overlap. The IZA Evaluation Dataset covers workers who became unemployed in 2007 and 2008 and follows them for up to three years. The IAB Job Vacancy Survey covers the period from 2000 to 2014. For the structural estimation of our model, we therefore restrict our attention to 2007-2011, the years for which information from both datasets is available.¹⁰

For the IZA Evaluation Dataset, we begin with the full panel of individuals who entered unemployment in 2007 and 2008, and include all survey waves. We restrict the sample to individuals observed in at least two survey waves, which ensures that the data cover more than the initial months of unemployment. We exclude observations with missing sector information, and workers with reported monthly income above the 98th percentile or below the 2nd percentile. We further exclude individuals younger than 18 and observations with reported unemployment benefits, numbers of monthly applications, or numbers of monthly vacancy referrals above the respective 98th percentiles. The resulting analysis sample consists of 6,121 individuals and 15,795 person-year observations.

Our initial sample from the IAB Job Vacancy Survey includes all firms surveyed between 2007 and 2011. We exclude firms that report having filled a vacancy despite reporting zero applicants. The resulting analysis sample consists of 36,349 firms.

Descriptives To provide an overview of the German labor market in our sample period, we report summary statistics on the search behavior of unemployed workers in Table 1 and on the hiring behavior of firms in Table 2.

Table 1 reports descriptive statistics for the IZA Evaluation Dataset sample. The summary statistics indicate that the average accepted wage of individuals who exit unemployment substantially exceeds the average monthly unemployment benefits claimed by unemployed individuals, by a factor greater than 1.7, suggesting a large employment premium. Unemployed individuals in our sample send an average of 6.9 monthly regular applications and receive 1.0 vacancy refer-

¹⁰It is worth noting that although our sample period contains the time of the global financial crisis, the German labor market proved surprisingly robust during the 2009 recession, with unemployment and wages remaining relatively stable (Burda and Hunt, 2011; Möller, 2010). This resilience has often been attributed to wage moderation in the years preceding the recession as well as to the extensive use of short-time work arrangements (Boeri and Bruecker, 2014).

als (VRs). Interestingly, individuals who receive at least one VR report, on average, 1.9 more applications than those who do not receive a VR, suggesting that there are complementarities between VRs and regular applications rather than crowding out.¹¹ Finally, individuals in our sample are on average 36 years old, and 47% are female.

Table 2 provides summary statistics on firms' hiring decisions from the IAB Job Vacancy Survey. Firms report an average of 5.5 hires during the 12 months preceding the survey interview. For the most recently filled vacancy, firms receive an average of 9.6 applications per month, and the average vacancy duration is 2.4 months.

Table 1: Summary statistics, IZA Evaluation Dataset

Variable	Mean	Std. Dev.
Monthly unemployment benefits	677	342
Accepted monthly wages	1211	626
# regular applications per month	6.94	7.62
# regular applications per month, if VRs=0	5.92	7.17
# regular applications per month, if VRs>0	7.83	7.88
# VRs per month	1.00	1.54
Age	35.74	10.46
Female	0.47	0.50
East German	0.32	0.47
Agriculture, forestry, and fishing	0.04	0.19
Manufacturing	0.26	0.44
Services	0.70	0.46

Notes: Summary statistics based on the 6,121 individuals in our IZA Evaluation Dataset analysis sample. Statistics on accepted wages are conditional on employment. Statistics on unemployment benefits, number of applications, and number of VRs are conditional on unemployment. Data on unemployment benefits and accepted wages are measured in Euros. Data on number of applications and number of VRs correspond to applications sent and VRs received during the respondent's most recent unemployment spell prior to the survey interview and are converted into monthly measures by dividing by spell duration. East German refers to residing in East Germany. Sector classifications are based on the first sector observed for a worker in the data.

¹¹The positive correlation between the number of applications and VRs, $\text{Cov}(\alpha, \iota) = 0.1$, points in the same direction.

Table 2: Summary statistics, IAB Job Vacancy Survey

Variable	Mean	Std. Dev.
# hires	5.53	27.40
# regular applications received per month (last vacancy)	9.55	22.40
Vacancy duration in month	2.42	2.12
East Germany	0.22	0.41
Agriculture, forestry, and fishing	0.03	0.17
Manufacturing	0.28	0.45
Services	0.69	0.46

Notes: Summary statistics based on the 36,349 firms in our IAB Job Vacancy Survey analysis sample. The number of hires refers to the 12 months preceding the survey interview. Data on applications received relate to the most recently filled vacancy at the surveyed firms and are converted into monthly measures by dividing by vacancy duration. East Germany refers to firms located in East Germany.

Figure 1 illustrates the distributions of the number of monthly applications sent and VRs received by workers in panels A and B, and the distributions of applications and VRs received by firms for the last vacancy they filled in panels C and D.

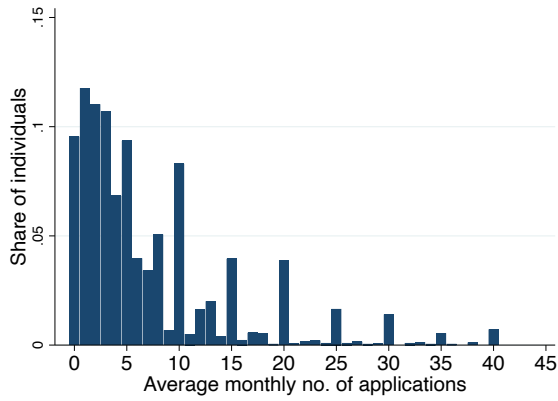
Panel A shows substantial bunching at round numbers in workers' reported numbers of applications, particularly for values above 10. In our sample, 6% of workers report having sent no applications. The median of the distribution is 8 applications and a small share of workers (less than 5%) report having sent out more than 40 applications.

Panel B shows that 44% of unemployed workers report not having received a VR in the month prior to the interview, while 17% report having received exactly one. The 95th percentile of the distribution is 8 VRs.

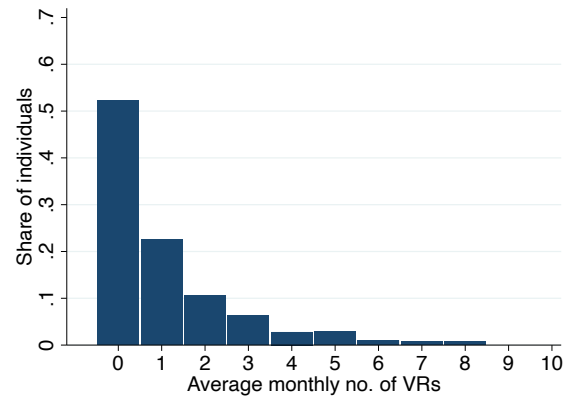
Panel C shows that 19% of firms report having received only one application for their most recent job opening. The median and the 90th percentile of the distribution are 4 and 26 applications, respectively. The distribution exhibits strong right skewness. 3.5% of firms reporting more than 40 applications for their most recent job opening.

Panel D shows that a large share of firms (31%) report having received no VR-applications. Extremely few firms (less than one percent) report having received more than 15 VR-applications for their most recent job opening. 42% report having received 1-5, 15% report 6-10 and 6% report having received 10-15 VR-applications.

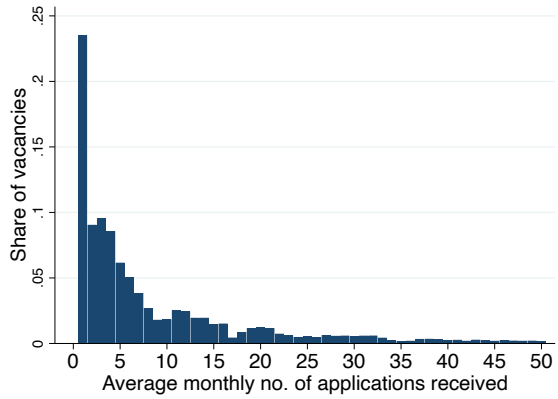
Figure 1: Empirical Distributions of Applications and VRs, Worker-side and Firm-side



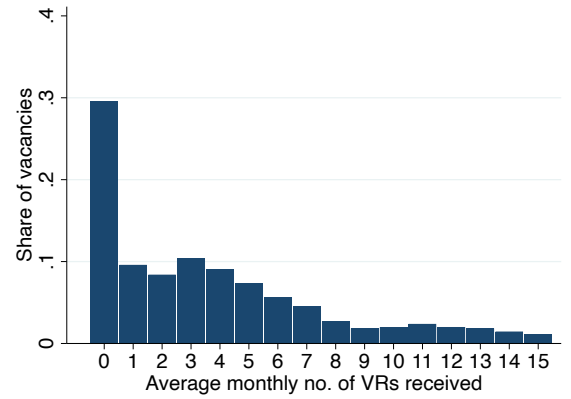
(a) Applications submitted by workers



(b) VRs received by workers



(c) Regular applications received by vacancies



(d) Applicants via VRs received by vacancies

Notes: The figure reports the empirical distributions of monthly applications submitted and VRs received by workers, and regular applications and applicants via VRs received by vacancies. Panel (a) reports the number of applications submitted per month, while Panel (b) reports the number of VRs received per month. Panel (c) reports the monthly number of applications received by vacancies, while Panel (d) reports the monthly number of applicants via VRs received by vacancies. Panels (a) and (b) are based on our analysis sample of 6,121 unemployed workers from the IZA Evaluation Dataset. Panels (c) and (d) are based on our analysis sample of 36,349 vacancies from the IAB Job Vacancy Survey. Horizontal axes report monthly counts and vertical axes report the corresponding shares of workers or vacancies.

5 Estimation

To obtain estimates of the structural parameters of the model, we proceed in three steps. First, a subset of parameters is set externally to match values from the existing literature and external data sources. Second, several parameters are fixed to match their empirical counterparts measured in the IZA Evaluation Dataset. Finally, the remaining parameters are estimated using the generalized method of moments (GMM). We estimate the model separately for four submarkets, indexed by $m \in \{1, \dots, 4\}$, defined by region (East vs. West Germany) and sector (services vs. manufacturing).¹² For tractability, we treat submarket affiliation as fixed when estimating the model, abstracting from worker transitions across submarkets.

5.1 Pre-set and directly estimated parameters

The parameters that are set externally are summarized in Table 3. We fix the monthly discount factor at $\beta = 0.997$. Monthly unemployment benefits, b_m , vary across submarkets and are set to the region-sector-specific averages observed in our IZA Evaluation Dataset sample. We set the maximum number of monthly applications to $A = 40$ and the maximum number of monthly VRs to $J = 8$, corresponding to the 98th percentiles of their respective empirical distributions.

A VR policy in submarket m is characterized by a probability distribution $p_{vr,m}^0 = (p_{vr,m,j}^0)_{j=0}^8$ where $p_{vr,m,j}^0$ denotes the probability that an unemployed worker in submarket m receives j VRs in a month. We set $p_{vr,m}^0$ to the corresponding region-sector-specific empirical distribution of monthly VRs.

Table 3: Externally set parameters

Parameter	Value	Source
Monthly discount factor, β	0.997	—
Monthly unemployment benefits, b_m	see Table B.1	IZA Evaluation Data
Max. number of monthly applications, A	40	IZA Evaluation Data, 98th percentile
Max. number of monthly VRs, J	8	IZA Evaluation Data, 98th percentile
VR distribution, $p_{vr,m}^0$	see Table B.2	IZA Evaluation Data

¹²We exclude the primary sector from the estimation, as only 4% of workers in our sample work in agriculture, forestry, and fishing.

5.2 Generalized method of moments estimation

For each submarket m , the remaining parameters are estimated using the generalized method of moments (GMM). These parameters are the vacancy creation cost, k_m , the match flow value, y_m , the parameter governing the effect of VRs on application costs, ϕ_m , and the job destruction rate, δ_m . We further assume that the application cost parameter, c_m , is log-normally distributed within each submarket,

$$\ln(c_m) \sim \mathcal{N}(\mu_m, \sigma_m),$$

and estimate μ_m and σ_m . We denote the vector of parameters for submarket m by

$$\theta_m = (k_m, y_m, \phi_m, \delta_m, \mu_m, \sigma_m).$$

For a given parameter vector θ_m , we solve the structural model using dynamic programming.¹³ Having solved the model, we compute a vector of model-implied moments, $g_m(\theta_m)$. The GMM estimator minimizes the distance between the vector of model-implied moments and its empirical counterpart, $\hat{g}_m$, separately for each submarket:

$$\hat{\theta}_m = \underset{\theta_m}{\operatorname{argmin}} (g_m(\theta_m) - \hat{g}_m)' \widehat{W}_m (g_m(\theta_m) - \hat{g}_m), \quad m = 1, \dots, 4, \quad (12)$$

where $\widehat{W}_m$ is a diagonal weighting matrix with elements given by the inverse variances of the empirical moments in $\hat{g}_m$.

For the estimation, in each submarket we target 14 empirical moments that can be approximated or computed directly as functions of the structural parameters. These moments include the shares for which the number of regular applications, α , lies in each of eight disjoint intervals (8 moments), the mean number of regular applications submitted by workers who received at least one VR ($\iota > 0$) and by those who received no VR ($\iota = 0$) (2 moments), and the mean accepted wage (1 moment). We further target the average number of applications received by firms per vacancy, the employer matching rate, and the worker matching rate (3 moments).

5.3 Parameter Estimates and Model Fit

The GMM parameter estimates for the four submarkets are presented in Table 4. The estimates indicate meaningful heterogeneity across submarkets. For example, estimated vacancy

¹³To implement the model numerically, we approximate the wage offer distribution by a piecewise linear cumulative distribution function, discretize the application cost distribution, and use numerical integration to approximate integrals.

posting costs range from 1,335 to 2,556 and mean match flow values from 1,135 to 1,434. At the same time, the estimated parameters of the application cost distribution are relatively similar across submarkets. Finally, we estimate $\phi_m < 0$ in all four submarkets, consistent with complementarities between VRs and regular applications.

Table 4: GMM parameter estimates

Parameter	West		East	
	Manufacturing	Services	Manufacturing	Services
Application cost location, μ_m	3.10	3.00	3.11	2.78
Application cost scale, σ_m	1.97	1.89	1.85	2.01
Vacancy posting cost, k_m	1867	2556	1335	1345
VR complementarity, ϕ_m	-1.19	-1.58	-0.84	-1.60
Mean match flow value, $\mu_{y,m}$	1434	1296	1350	1135
S.d. match flow value, $\sigma_{y,m}$	213	165	174	130

Notes: The table reports GMM parameter estimates for each of the four region-sector submarkets. The parameters μ_m and σ_m characterize the distribution of application costs, $\ln(c) \sim \mathcal{N}(\mu_m, \sigma_m^2)$. The parameters $\mu_{y,m}$ and $\sigma_{y,m}$ denote the mean and standard deviation of the match flow value distribution. The parameter ϕ_m governs how receiving VRs affects application costs.

Table 5: Model fit by sector and region

Sector	Moment	West		East	
		Data	Model	Data	Model
Manufacturing	Mean applications received per vacancy	9.14	10.40	5.93	7.02
	Mean applications submitted, no referral	5.59	5.81	5.84	5.30
	Mean applications submitted, with referral	8.36	7.63	6.49	6.12
	Job-finding rate	0.26	0.27	0.28	0.33
	Vacancy-filling rate	0.35	0.35	0.38	0.36
	Mean accepted wage	1374	1371	1271	1272
Services	Mean applications received per vacancy	10.63	13.60	6.69	9.78
	Mean applications submitted, no referral	5.81	5.90	6.42	6.15
	Mean applications submitted, with referral	8.77	8.40	8.98	8.14
	Job-finding rate	0.25	0.26	0.24	0.29
	Vacancy-filling rate	0.44	0.43	0.42	0.36
	Mean accepted wage	1213	1212	1087	1085

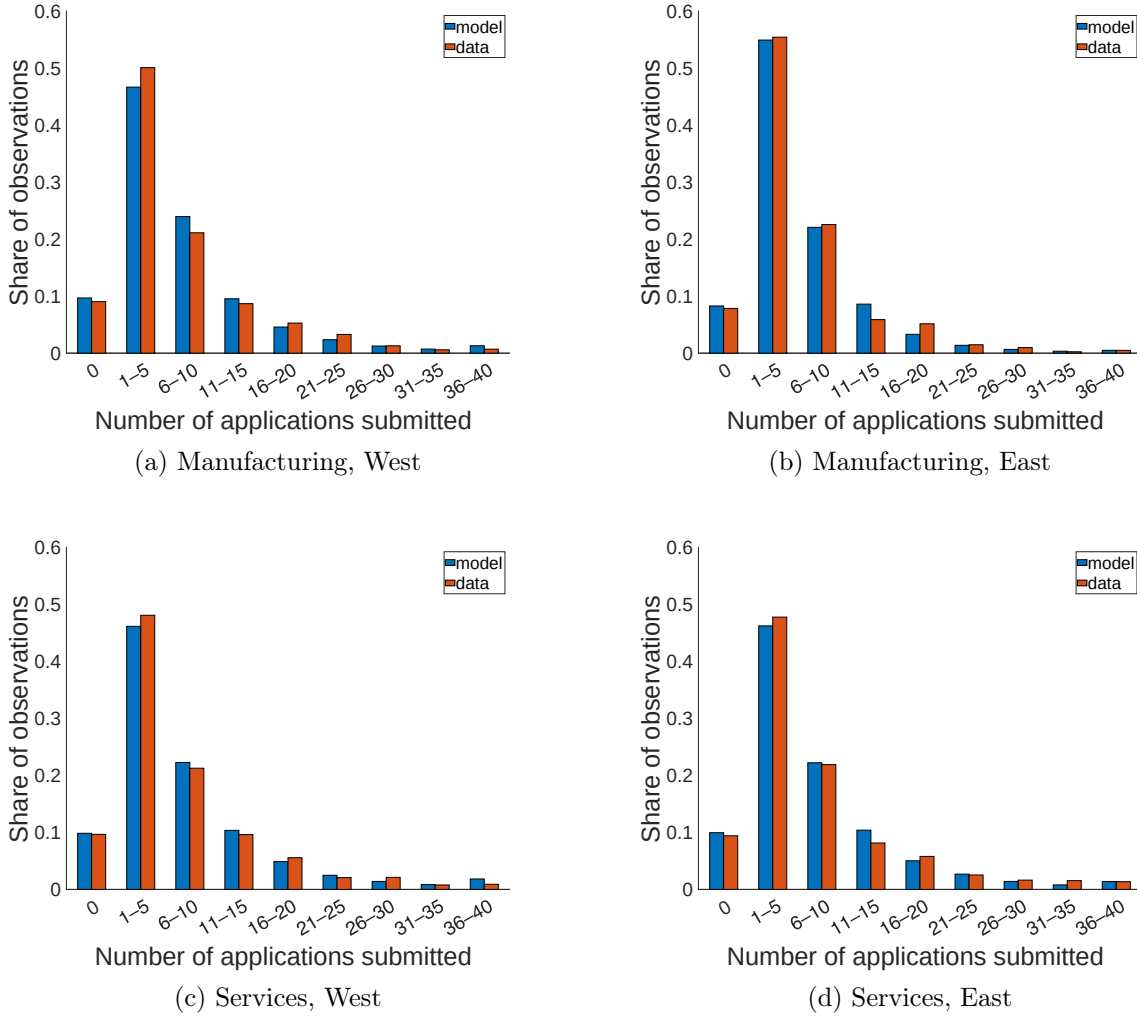
Notes: The table compares empirical moments and their model counterparts across the four labor markets defined by sector and region.

To assess the model fit, Table 5 and Figure 2 compare targeted empirical moments with their model-implied counterparts evaluated at the estimated parameter values for each of the four submarkets. The estimated model provides a good fit for most of the targeted moments reported in Table 5. In particular, it closely matches mean accepted wages, vacancy-filling rates,

and the number of regular applications submitted by unemployed workers. It also captures that workers who receive VRs submit more regular applications than workers who do not receive VRs. The number of applications received per vacancy is somewhat overpredicted, especially in East Germany, where the empirical values are low relative to vacancy-filling rates.

Figure 2 further shows that the model closely matches the distribution of regular applications submitted by unemployed workers in each submarket.

Figure 2: Model fit: Distribution of applications



Notes: The figure compares the empirical and model-implied distributions of the number of regular applications submitted by unemployed workers. The model-implied distributions are computed at the estimated parameter values for each submarket.

6 Counterfactual Simulations

To inform how VRs affect labor market outcomes and how their effects compare to alternative labor market policies, we consider several sets of counterfactual policy scenarios. First, we con-

sider counterfactual scenarios in which VR rates are increased or reduced. Second, we compare VRs to other active labor market policies. Third, we consider targeting VRs toward individuals with high search costs.

6.1 The impact of VRs on aggregate labor market outcomes

As described in Section 5, a VR policy in submarket m is characterized by the probability distribution $p_{vr,m} = (p_{vr,m,j})_{j=0}^8$, where $p_{vr,m,j}$ denotes the probability that an unemployed worker receives j VRs in a month. The status-quo policy, denoted by $p_{vr,m}^0$, corresponds to the submarket-specific VR distributions observed in the data.

To reduce the dimensionality of the policy space we consider a parameterization that scales up or scales down the status-quo VR policy. In particular, for each submarket m , we define

$$p_{vr,m,j}(\kappa) = \kappa p_{vr,m,j}^0, \quad j = 1, \dots, 8, \quad (13)$$

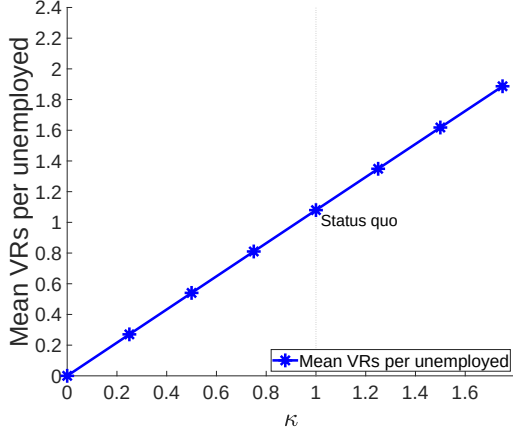
and set the residual probability of receiving no VRs to

$$p_{vr,m,0}(\kappa) = 1 - \kappa (1 - p_{vr,m,0}^0). \quad (14)$$

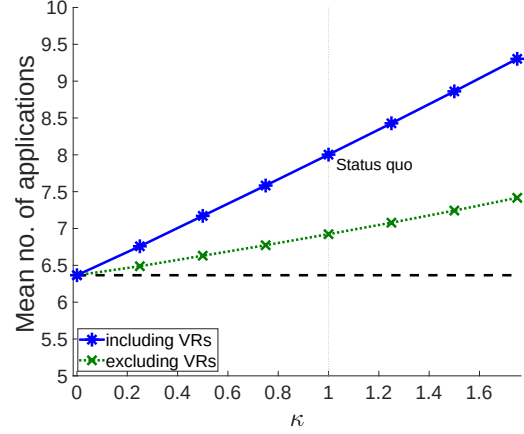
This parameterization nests a policy with no VRs for $\kappa = 0$ and the status-quo policy for $\kappa = 1$. Moreover, the mean number of VRs received scales linearly with κ , both within each submarket and in the aggregate, allowing us to scale down ($\kappa < 1$) or scale up ($\kappa > 1$) VR provision relative to the status quo. We solve the model in steady state for a range of values of κ , holding all other model parameters fixed. To report economy-wide effects, we aggregate outcomes across the four submarkets.

Figure 3 shows that higher VR rates increase the total volume of applications and VRs in the labor market. As unemployed workers receive more VRs, they also submit more regular applications, reflecting the complementarities between VRs and regular applications implied by our negative estimates of ϕ_m . The resulting increase in contacts between workers and vacancies raises the average queue length and thus congestion at vacancies. Despite this rise in congestion, worker matching rates increase substantially. For example, increasing the mean number of VRs per unemployed worker from around 1.1 under the status-quo policy to around 1.6 raises the monthly worker matching rate from about 27.5% to 29%. Finally, higher VR rates are associated with increased labor market tightness, while posted and realized wages increase modestly.

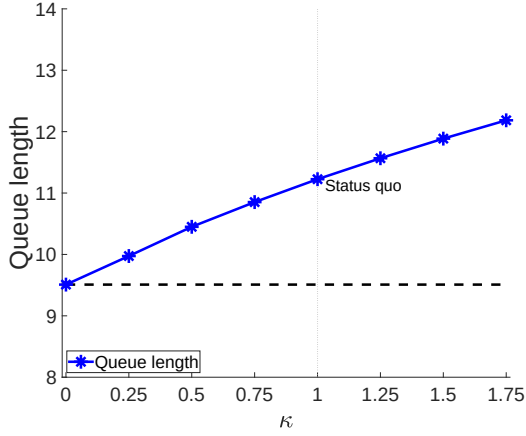
Figure 3: Equilibrium effects of varying VR policy intensity



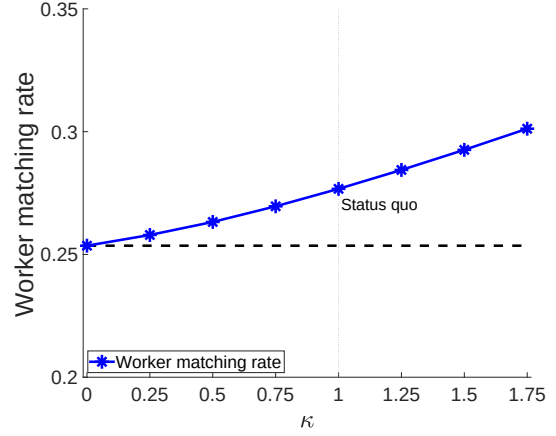
(a) Mean VRs received per unemployed



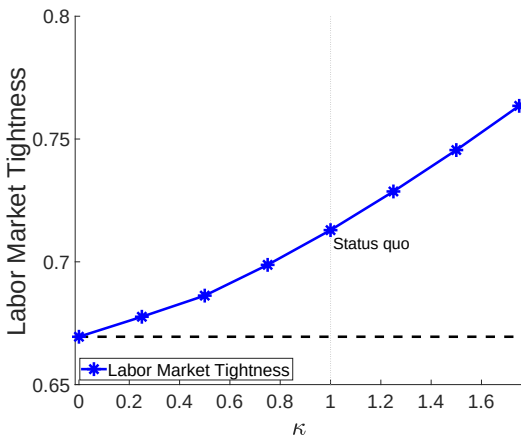
(b) Mean applications submitted



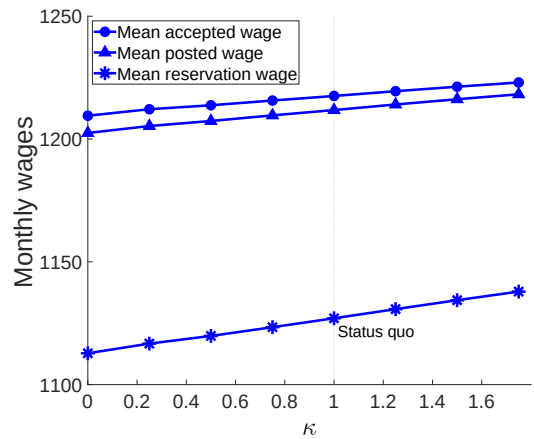
(c) Queue length



(d) Worker matching rates



(e) Labor market tightness



(f) Wages

Notes: All outcomes are aggregated across the four submarkets. The policy parameter κ refers to the parameterization of the VR policy given by Equations 13 and 14. Panel (a) reports the average number of VRs received per unemployed worker per month. Panel (b) reports the average number of applications submitted and VRs received per unemployed worker per month. Panel (c) reports the average number of applications and VRs per vacancy. Panel (d) reports the monthly worker matching rate. Panel (e) reports labor market tightness, defined as the aggregate vacancy rate divided by the aggregate unemployment rate. Panel (f) reports monthly wages in euros.

6.2 Quantifying the equilibrium effects of VRs

To quantify the equilibrium effects of VRs, we decompose the aggregate effect of rolling out a given VR policy on the worker matching rate. Consider a VR policy given by $\kappa = \tilde{\kappa}$. Suppose that a fraction ρ of unemployed workers is treated with this policy, while the remaining fraction $1 - \rho$ receive no VRs. We refer to the former as treated workers and the latter as untreated workers.

Let $m_{W|\kappa=\tilde{\kappa}}(\rho)$ denote the matching rate of treated workers and $m_{W|\kappa=0}(\rho)$ the matching rate of untreated workers when a fraction ρ of unemployed workers is treated. Both matching rates depend on ρ because the scale of the policy affects equilibrium labor market conditions.

The aggregate worker matching rate can then be written as a weighted average of the matching rates of treated and untreated workers

$$m_W(\rho) = \rho m_{W|\kappa=\tilde{\kappa}}(\rho) + (1 - \rho) m_{W|\kappa=0}(\rho). \quad (15)$$

Using this representation, the effect of a policy reform that increases the share of treated workers from ρ to ρ' can be decomposed as

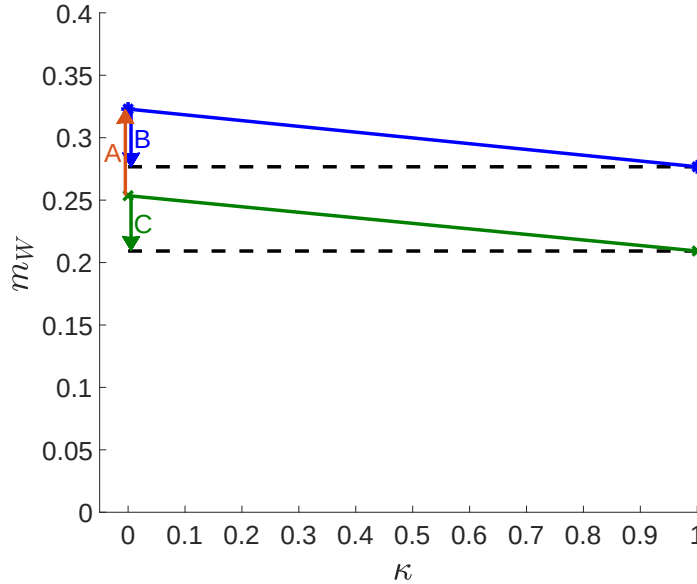
$$\begin{aligned} m_W(\rho') - m_W(\rho) &= (\rho' - \rho) \underbrace{\left[m_{W|\kappa=\tilde{\kappa}}(\rho) - m_{W|\kappa=0}(\rho) \right]}_{\mathbf{A}(\rho)} \\ &\quad + \rho' \underbrace{\left[m_{W|\kappa=\tilde{\kappa}}(\rho') - m_{W|\kappa=\tilde{\kappa}}(\rho) \right]}_{\mathbf{B}(\rho, \rho')} \\ &\quad + (1 - \rho') \underbrace{\left[m_{W|\kappa=0}(\rho') - m_{W|\kappa=0}(\rho) \right]}_{\mathbf{C}(\rho, \rho')}. \end{aligned} \quad (16)$$

The first component, $\mathbf{A}(\rho)$, captures the direct treatment effect of assigning the VR policy to additional unemployed workers, holding equilibrium conditions fixed. It reflects the difference in matching rates between a worker assigned the VR policy and a worker assigned the no-VR policy, evaluated at the pre-reform equilibrium. The second and third components, $\mathbf{B}(\rho, \rho')$ and $\mathbf{C}(\rho, \rho')$ capture equilibrium effects on treated and untreated workers, respectively. These effects arise because expanding the policy changes application behavior, congestion in the hiring process, and firms' vacancy posting decisions, thereby affecting matching probabilities even for workers whose own treatment status is held fixed.¹⁴

¹⁴In the language of the treatment evaluation literature (see, e.g., Imbens and Angrist (1994) and Heckman and Vytlačil (2007)), the first component corresponds to a treatment effect defined under the assumption that equilibrium spillovers are absent (the standard Stable Unit Treatment Value Assumption, SUTVA). The second and third components capture equilibrium effects on treated and untreated individuals, respectively, which lie

Figure 4 applies this decomposition to the status-quo VR policy, $\tilde{\kappa} = 1$, starting from an economy in which no workers are treated with this policy ($\rho = 0$). The figure plots the matching rates of treated and untreated workers as the share of workers treated with the status-quo policy increases to $\rho' = 1$. We find that fully rolling out the policy raises the aggregate worker matching rate by 2.3 percentage points. The individual treatment effect is positive ($\mathbf{A} > 0$), while the equilibrium effects on both treated and untreated workers are negative ($\mathbf{B} < 0$ and $\mathbf{C} < 0$). Our decomposition shows that absent equilibrium effects, the impact of the policy on the worker matching rate would be 6.9 percentage points. Thus, absent equilibrium effects the impact of the policy roll out would be roughly three times as large.

Figure 4: Individual treatment and equilibrium effects of vacancy referrals



Notes: The figure illustrates the decomposition of the effect of vacancy referrals on the worker matching rate. **A** denotes the individual treatment effect of assigning the status-quo VR policy to a single unemployed worker, holding equilibrium conditions fixed. **B** and **C** denote the equilibrium effects of rolling out the policy to the entire population on treated and untreated unemployed workers, respectively.

6.3 Comparison with alternative labor market policies

To put the effects of VRs into perspective, we next compare them to those of two alternative active labor market policies. Starting from an equilibrium without VRs ($\kappa = 0$), we consider a 10% reduction in unemployment benefits, b , and a 10% reduction in application costs, c , for all workers, which mimics policies that provide job search assistance (e.g., counselling). We compare these counterfactuals to introducing the status-quo VR policy ($\kappa = 1$).

outside the scope of standard treatment evaluation frameworks.

Table 6 shows that introducing the status-quo VR policy and reducing unemployment benefits have qualitatively similar effects on most labor market outcomes. The exceptions are reservation, posted, and realized wages, which decline modestly when unemployment benefits are reduced, as lower benefits reduce the value of search for unemployed workers.

The effects of introducing the status-quo VR policy and reducing application costs are qualitatively similar across all outcomes, suggesting that VRs operate similarly to other active labor market policies that facilitate job search. Quantitatively, the VR policy generates a somewhat larger increase in the worker matching rate and a larger decline in unemployment than the reductions in either benefits or application costs, for the magnitudes of the policy changes considered here.

Table 6: Counterfactual policy simulations

Outcome	No VRs (baseline)	Status quo VRs	Lowering benefits	Lowering appli- cation costs
Mean applications submitted	6.37	6.92 (+9%)	6.58 (+3%)	7.96 (+25%)
Mean VRs received by workers	0.00	1.08 (-)	0.00 (-)	0.00 (-)
Job-finding rate	0.25	0.28 (+9%)	0.27 (+6%)	0.26 (+3%)
Unemployment rate	0.102	0.095 (-8%)	0.097 (-5%)	0.100 (-2%)
Mean applications received by vacancies	9.51	9.71 (+2%)	9.18 (-3%)	11.57 (+22%)
Mean VRs received by vacancies	0.00	1.51 (-)	0.00 (-)	0.00 (-)
Vacancy-filling rate	0.38	0.39 (+2%)	0.37 (-1%)	0.38 (+0%)
Vacancies	0.069	0.068 (-2%)	0.070 (+2%)	0.069 (+0%)
Labor market tightness	0.67	0.71 (+6%)	0.72 (+7%)	0.69 (+3%)
Mean posted wage	1202	1212 (+1%)	1198 (-0%)	1202 (+0%)
Mean reservation wage	1113	1127 (+1%)	1104 (-1%)	1110 (+0%)
Mean realized wage	1209	1218 (+1%)	1206 (-0%)	1209 (+0%)

Notes: Job-finding and vacancy-filling rates and all application and VR measures are monthly. Wages are monthly and reported in euros. “No VRs” sets the VR policy parameter to $\kappa = 0$, while “Status quo VRs” sets $\kappa = 1$, corresponding to the VR distribution observed in the data. “Lowering benefits” reduces unemployment benefits, b , by 10%, and “Lowering application costs” reduces the application-cost parameter, c , by 10%. Both counterfactuals are implemented in the absence of VRs (i.e., maintaining $\kappa = 0$). Percentage changes relative to the no-VR baseline are reported in parentheses.

6.4 Targeting vacancy referrals

Next, we investigate whether the effectiveness of VRs can be increased by targeting them toward workers who would otherwise have low odds of being matched. Specifically, we consider a counterfactual that redistributes VRs from workers with low application costs to workers with high application costs, while leaving the aggregate mass of VRs in the economy.

Formally, define an application cost cutoff $\bar{c}$ and let s denote the share of unemployed workers with application costs above the cutoff. Let κ_h and κ_l denote the factors by which VR rates are

scaled for workers above and below the cutoff, respectively. We then choose these scaling factors such that

$$(1 - s)p_{vr,m}(\kappa_l) + sp_{vr,m}(\kappa_h) = p_{vr,m}^0. \quad (17)$$

Given the linear parameterization of the VR policy in Equations 13 and 14, this condition is satisfied whenever

$$(1 - s)\kappa_l + s\kappa_h = 1. \quad (18)$$

As a simple example, consider a counterfactual in which $\bar{c}$ is the median of the application-cost distribution, so that $s = 0.5$. As a counterfactual exercise, we increase VR rates by 50% for workers above the median ($\kappa_h = 1.5$) and reduce them by 50% for workers below the median ($\kappa_l = 0.5$). Thus, the counterfactual targets VRs toward workers who would otherwise submit fewer applications without changing the aggregate VR distribution.

Table 7: Counterfactual redistribution of vacancy referrals

Outcome	Status quo VRs (baseline)	Redistributed VRs
Mean applications submitted	6.92	6.45 (-7%)
Mean VRs received by workers	1.08	1.08 (+0%)
Job-finding rate	0.28	0.32 (+15%)
Unemployment rate	0.095	0.084 (-12%)
Mean applications received by vacancies	9.71	8.10 (-17%)
Mean VRs received by vacancies	1.51	1.35 (-11%)
Vacancy-filling rate	0.39	0.40 (+3%)
Vacancies	0.068	0.067 (-1%)
Labor market tightness	0.71	0.80 (+12%)
Mean posted wage	1212	1223 (+1%)
Mean reservation wage	1127	1146 (+2%)
Mean realized wage	1218	1226 (+1%)
Share of unemployed with 0 applications + VRs	0.08	0.02 (-80%)
Share of unemployed with at most 1 application + VRs	0.18	0.04 (-78%)
Share of unemployed with at most 3 applications + VRs	0.35	0.23 (-36%)

Notes: Job-finding and vacancy-filling rates and all application and VR measures are monthly. Wages are monthly and reported in euros. “Status quo VRs” sets the VR policy parameter to $\kappa = 1$ for all workers. “Redistributed VRs” increases VR rates by 50% for workers above the median of the application-cost distribution ($\kappa_h = 1.5$) and reduces VR rates by 50% for workers below the median ($\kappa_l = 0.5$). This leaves the aggregate VR distribution unchanged. Percentage changes relative to the status-quo VR policy are reported in parentheses.

Table 7 shows that targeting VRs in this way substantially increases their effectiveness. Although unemployed workers submit fewer applications on average and vacancies consequently receive fewer applications and VRs, the worker matching rate increases by 15% and unemploy-

ment falls by 12%. The vacancy-filling rate also increases, while wages rise modestly.

The distribution of worker search activity points to the mechanism behind these effects. Under the considered counterfactual, targeting VRs sharply reduces the share of unemployed workers with very few applications and VRs: the share with no applications or VRs falls by 80%, while the share with at most one falls by 78%, suggesting the policy reduces unemployment by targeting VRs to workers who otherwise have low odds of being matched.

7 Conclusion

In this paper, we study the equilibrium effects of vacancy referrals (VRs). We develop and estimate an equilibrium search model in which unemployed workers choose how many regular applications to submit and may additionally receive VRs from the public employment service, while firms endogenously choose vacancy posting and wages. We estimate the model separately across four labor market submarkets defined by region and sector.

Our counterfactual analysis shows that expanding VRs increases the total volume of applications and referrals in the labor market and raises worker matching rates. At the same time, the additional search activity increases congestion at vacancies, generating negative equilibrium spillovers. Higher VR rates nevertheless reduce unemployment and increase vacancy posting, while wages increase modestly. We also find that the effects of VRs are qualitatively similar to those of other policies that facilitate job search, such as reductions in application costs, and that targeting VRs toward workers with high search costs can improve their effectiveness. Our results highlight the quantitative importance of accounting for equilibrium effects when evaluating active labor market policies. Thus, small-scale quasi-experimental evaluations that identify individual-level effects of VRs on reemployment probabilities may substantially overstate their effects when implemented at scale.

An interesting extension would be to incorporate on-the-job search, which may generate additional search activity and larger applicant pools competing for vacancies. Another natural direction for future research would be to allow firms to recruit through channels other than officially posted vacancies. More broadly, incorporating additional sources of worker and firm heterogeneity could shed light on how the equilibrium effects of VRs vary across different types of workers and employers.

References

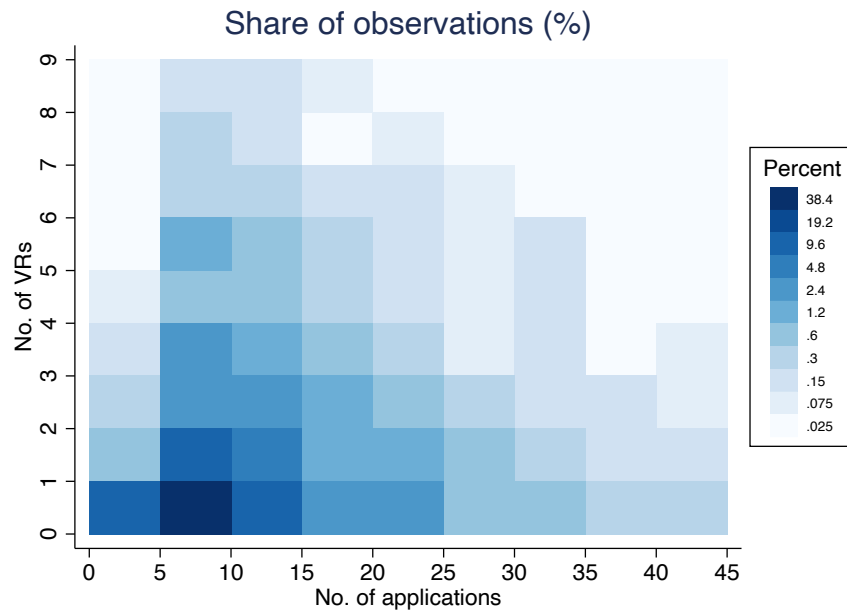
- Abbring, Jaap and James Heckman (2007). “Econometric Evaluation of Social Programs, Part III: Distributional Treatment Effects, Dynamic Treatment Effects, Dynamic Discrete Choice, and General Equilibrium Policy Evaluation”. In: *Handbook of Econometrics*. Ed. by J.J. Heckman and E.E. Leamer. 1st ed. Vol. 6B. Elsevier. Chap. 72.
- Albrecht, James, Pieter A. Gautier, and Susan Vroman (2006). “Equilibrium Directed Search with Multiple Applications”. In: *The Review of Economic Studies* 73.4, pp. 869–891. ISSN: 00346527, 1467937X.
- Albrecht, James, Serene Tan, Pieter Gautier, and Susan Vroman (2004). “Matching with multiple applications revisited”. In: *Economics Letters* 84.3, pp. 311–314. ISSN: 0165-1765. DOI: <https://doi.org/10.1016/j.econlet.2003.12.024>.
- Arni, Patrick, Marco Caliendo, Steffen Künn, and Klaus F. Zimmermann (2014). “The IZA evaluation dataset survey: a scientific use file”. In: *IZA Journal of European Labor Studies* 3.1, p. 6. DOI: 10.1186/2193-9012-3-6.
- Boeri, Tito and Herbert Bruecker (Aug. 2014). “Short-time work benefits revisited: some lessons from the Great Recession”. In: *Economic Policy* 26.68, pp. 697–765. ISSN: 0266-4658. DOI: 10.1111/j.1468-0327.2011.271.x. eprint: <https://academic.oup.com/economicpolicy/article-pdf/26/68/697/9571060/economicpolicy26-0697.pdf>.
- Bollens, Joost and Bart Cockx (2017). “Effectiveness of a job vacancy referral scheme”. In: *IZA Journal of Labor Policy* 6.1, p. 15. ISSN: 2193-9004. DOI: 10.1186/s40173-017-0094-0.
- Burda, Michael C. and Jennifer Hunt (2011). “What Explains the German Labor Market Miracle in the Great Recession”. In: *Brookings Papers on Economic Activity* 42.1 (Spring, pp. 273–335. DOI: None.
- Caliendo, Marco, Armin Falk, Lutz C. Kaiser, Hilmar Schneider, Arne Uhlenhorff, Gerard van den Berg, and Klaus F. Zimmermann (2011). “The IZA Evaluation Dataset: towards evidence-based labor policy making”. In: *International Journal of Manpower* 32.7, pp. 731–752. DOI: 10.1108/01437721111174730. eprint: <https://doi.org/10.1108/01437721111174730>.
- Crépon, Bruno, Esther Duflo, Marc Gurgand, Roland Rathelot, and Philippe Zamora (Apr. 2013). “Do Labor Market Policies have Displacement Effects? Evidence from a Clustered Randomized Experiment *”. In: *The Quarterly Journal of Economics* 128.2, pp. 531–580.
- Engström, Per, Patrik Hesselius, and Bertil Holmlund (2012). “Vacancy Referrals, Job Search, and the Duration of Unemployment: A Randomized Experiment”. In: *LABOUR* 26.4, pp. 419–435.
- Fougère, Denis, Jacqueline Pradel, and Muriel Roger (2009). “Does the public employment service affect search effort and outcomes?” In: *European Economic Review* 53.7, pp. 846–869.
- Gautier, Pieter A., José L. Moraga-González, and Ronald P. Wolthoff (2016). “Search costs and efficiency: Do unemployed workers search enough?” In: *European Economic Review* 84. European Labor Market Issues, pp. 123–139. ISSN: 0014-2921. DOI: <https://doi.org/10.1016/j.euroecorev.2015.04.001>.
- Gautier, Pieter, Paul Muller, Bas van der Klaauw, Michael Rosholm, and Michael Svarer (2018). “Estimating Equilibrium Effects of Job Search Assistance”. In: *Journal of Labor Economics* 36.4, pp. 1073–1125.

- Grubb, David (Jan. 2001). “Eligibility Criteria for Unemployment Benefits”. In: *OECD Economic Studies (Paris)* 31.
- Heckman, James J. and Edward J. Vytlacil (2007). “Econometric Evaluation of Social Programs, Part I: Causal Models, Structural Models and Econometric Policy Evaluation”. In: *Handbook of Econometrics*. Ed. by James J. Heckman and Edward E. Leamer. Vol. 6, Part 2. Elsevier. Chap. 70, pp. 4779–4874.
- Imbens, Guido W. and Joshua D. Angrist (1994). “Identification and Estimation of Local Average Treatment Effects”. In: *Econometrica* 62.2, pp. 467–475.
- Immervoll, Herwig and Carlo Knotz (2018). “How demanding are activation requirements for jobseekers”. In: *OECD Social, Employment and Migration Working Papers* 215.
- Kubis Alexander; Moczall, Andreas; Rebien Martina (2018). “The IAB Job Vacancy Survey. Establishment survey on labour demand and recruitment processes Waves 2000 to 2015 and subsequent quarters since 2006”. In: *FDZ-Datenreport, 05/2018 (en)*, Nuremberg.
- Lehmann, Tobias, Camille Terrier, and Rafael Lalive (2023). “Costs and Benefits of Congestion in Two-Sided Markets: Evidence from the Dating Market”. In: 964. DOI: **None**.
- Möller, Joachim (2010). “The German labor market response in the world recession : de-mystifying a miracle”. In: *Zeitschrift für ArbeitsmarktForschung - Journal for Labour Market Research* 42.4, pp. 325–336. DOI: 10.1007/s12651-009-0026-6.
- Petrongolo, Barbara and Christopher A. Pissarides (2001). “Looking into the Black Box: A Survey of the Matching Function”. In: *Journal of Economic Literature* 39.2, pp. 390–431. DOI: 10.1257/je1.39.2.390.
- Rogerson, Richard, Robert Shimer, and Randall Wright (2005). “Search-Theoretic Models of the Labor Market: A Survey”. In: *Journal of Economic Literature* 43.4, pp. 959–988. DOI: 10.1257/002205105775362014.
- van den Berg, Gerard J., Hanno Foerster, and Arne Uhlenborff (2019). “A Structural Analysis of Vacancy Referrals with Imperfect Monitoring and Sickness”. In: *working paper*.
- van den Berg, Gerard J., Barbara Hofmann, and Arne Uhlenborff (2014). “The Role of Sickness in the Evaluation of Job Search Assistance and Sanctions”.

Appendix

A Additional Figures

Figure A.1: Joint distribution of applications and vacancy referrals



Notes: The figure displays the empirical joint distribution of the number of regular applications submitted and the number of vacancy referrals (VRs) received by unemployed workers in the IZA Evaluation Dataset. Darker cells indicate a higher share of workers with the corresponding combination of applications and VRs.

B Additional Tables

Table B.1: Monthly unemployment benefits by submarket

	Manufacturing	Services
West	753	684
East	642	619

Notes: The table reports average monthly unemployment insurance benefits, in euros, among unemployed workers in each of the four region-sector submarkets. These values are used to set the unemployment benefit parameter, b_m , in the model.

Table B.2: Status-quo vacancy referral policies by submarket

Number of monthly VRs	West		East	
	Manufacturing	Services	Manufacturing	Services
0	0.462	0.504	0.579	0.563
1	0.223	0.235	0.202	0.219
2	0.106	0.110	0.091	0.112
3	0.082	0.067	0.052	0.054
4	0.041	0.029	0.027	0.023
5	0.049	0.028	0.027	0.019
6	0.016	0.013	0.007	0.003
7	0.014	0.007	0.005	0.004
8	0.008	0.008	0.010	0.004
Mean monthly VRs	1.25	1.08	0.93	0.87

Notes: The table reports the empirical distribution of the number of monthly vacancy referrals received by unemployed workers in each of the four region-sector submarkets. These distributions constitute the status-quo VR policies, $p_{vr,m}^0$, used in the model.